Group 6: Mobile Usage

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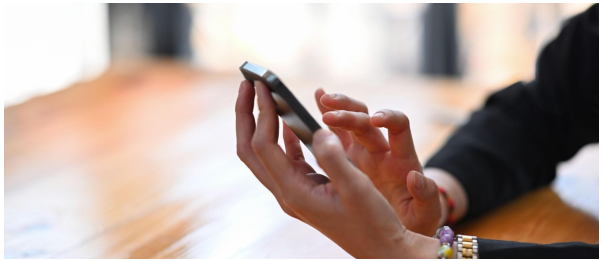
Domain

Mobile Usage Behavior

- Smartphone screen time and app usage
- Digital wellbeing and productivity
- Distraction vs. healthy habits
- Data-driven self-improvement



Goal: Help users understand and improve their mobile habits

Problem and Solution

Problems: Many people do not know if their phone usage is healthy or too much. Screen time numbers alone don’t clearly show if someone is being productive or distracted, and they do not give advice.

Solutions:This project uses different prompt techniques to study mobile phone usage. The model can classify user behavior, detect feelings, create simple dashboards, and give personal advice. It shows how clear prompts can turn simple phone data into helpful information.

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Scenario Overview

1. **Time Control Usage Chatbot**
   This scenario uses a system prompt to create a digital wellbeing chatbot that answers user questions and provides personalized guidance on healthy mobile usage.

2. **Mobile Usage Dashboard**
   This scenario generates structured HTML to display mobile usage statistics and visualize screen time and app activity in Colab

3. **Personalized Usage**
   This scenario uses multi-step prompting to analyze user behavior and generate personalized digital wellbeing recommendations.

4. **Screen Time Sentiment Analysis**
   This scenario applies labeled examples to classify user feedback as positive, neutral, or negative and improve sentiment analysis accuracy.

Scenario 1: Time Control Usage Chatbot

Technique: Zero-Shot Prompting + Chatbot Design

- Interactive digital wellbeing assistant
- Provides personalized usage advice
- Systems role defines supportive tone
- Refined prompt to reduce generic replies

Prompt Design

- Defined model as wellbeing expert
- Limited response length
- Enforced concise formating

Model Parameters

- Temperature: 0.7
- Max Tokens: 150

Why These Values?

- Higher temperature produces natural, engaging responses
- Token limit keeps answers concises

📱 Mobile Usage Digital Wellbeing Chatbot

Ask questions about your mobile usage and get personalized digital wellbeing advice.

Chatbot

I spend 6 hours daily on Instagram. Is that unhealthy?

Yes, spending 6 hours a day on Instagram is generally unhealthy, as it can affect your productivity, sleep, and mental health. It's important to balance social media use with other activities.

If you'd like, I can suggest practical steps to reduce your Instagram time and improve your digital wellbeing. Would you like some tips?

Scenario 2: Mobile Usage HTML Dashboard

Technique: HTML Generation

- Converts usage data into dashboards
- Generates HTML + inline CSS
- Enforces formatting rules
- Displays output on Colab

Prompt Design

- Defined strict HTML structure
- Restricted output format
- Refined prompt to reduce errors

Model Parameters

Temperature 0.3

Max output tokens: 800

Why These Values?

- Low temperature ensures structured output
- High token limit supports complete HTML generation

```python
from openai import OpenAI
from IPython.display import display, HTML
import os

# Use your existing API key
client = OpenAI()

# 📊 Sample mobile usage data
usage_data = """
Total Screen Time: 7 hours
Instagram: 3 hours
TikTok: 2 hours
YouTube: 1 hour
Notes: 0.5 hours
Calendar: 0.5 hours
"""
```

```python
# 🎯 HTML Generation Prompt
prompt = f"""
You are a web designer.

Generate a clean and modern HTML dashboard to display the following mobile usage statistics.

Requirements:
- Use only HTML and inline CSS.
- Include a title: "Mobile Usage Dashboard"
- Display total screen time clearly.
- Show app usage in a styled table.
- Use Dark Bleu colors and simple Grey layout.
- Output HTML code only. No explanations.

Mobile Usage Data:
{usage_data}
"""

response = client.responses.create(
    model="gpt-4.1-mini",
    input=prompt,
    temperature=0.3,   # Lower temperature for structured output
    max_output_tokens=800
)

# Render the HTML in Colab
display(HTML(response.output_text))
```

Mobile Usage Dashboard

7 hours

Total Screen Time

App	Usage
Instagram	3 hours
TikTok	2 hours
YouTube	1 hour

Scenario 3: Personalized Usage Prompt Chaining

Technique: Prompt Chaining

- Breaks complex task into two steps
- Improves reasoning accuracy
- Reduces generic recommendations

Process

- Step 1: Analyzes user behavior
- Step 2: Generated personalized advice

Prompt Design

- Separated analysis and recommendation
- Refined prompts to improve clarity
- Reduced reasoning to errors

Model Parameters

Step 1: Analysis

- Temperature: 0.6
- Max Tokens: 120

Step 2: Recommendations

- Temperature: 0.7
- Max tokens 200

Why These Values?

- Moderate temperatures ensures balanced reasoning in analysis
- Different limits optimize each task

Prompt Chaining Demo: Personalized Mobile Usage

user_input_data

Facebook: 1 Hour
Camera: 5 Hours
Messages: 3 Hours

Clear

Submit

output

User Classification:
Entertainment-focused (mostly social media and entertainment apps)

Recommendations:
1. Set specific time limits for social media and entertainment apps to avoid excessive use.
2. Schedule regular breaks to disconnect from digital devices and engage in offline activities.
3. Prioritize quality interactions with friends and family in person to balance online social interactions.

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Prompt Chaining Demo: Personalized Mobile Usage

user_input_data

Notes: .5 Hours
Messages: 2 Hours
Temple Run: 4 Hours

Clear

Submit

output

User Classification:
Entertainment-focused

The user spends a significant amount of time on Temple Run, indicating a focus on entertainment and gaming rather than productivity.

Recommendations:
1. Set time limits for gaming sessions to prevent excessive screen time.
2. Explore other types of entertainment outside of gaming to diversify digital activities.
3. Incorporate physical activities or hobbies into daily routines to balance screen time with offline experiences.

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Use via API · Built with Gradio · Settings

Scenario 4: Screen Time Sentiment Analysis

Technique: Few-shot Prompting

- Classifies feedback as positive, neutral, or negative
- Uses labeled examples to guide behavior
- Improves consistency and accuracy

Prompt Design

- Selected clear sentiment examples
- Refines labels to reduce ambiguity
- Balanced positive and negative samples

Parameters

- Temperature= 0.2
- Max Tokens: 16

Why These Values?

- Low temperature ensures consistent classification
- Low token limit prevents unnecessary output

```python
# Example user comment
user_comment = "I feel stressed when I see how much time I spend on social media."

# Few-Shot Prompt
prompt = f"""
You are a sentiment classifier specialized in mobile usage behavior.

Classify the emotional tone of the user's reflection as:
Positive
Neutral
Negative

Only output one word: Positive, Neutral, or Negative.

Examples:

Comment: "I reduced my screen time this week and I feel more productive."
Sentiment: Positive

Comment: "My screen time stayed the same as usual."
Sentiment: Neutral

Comment: "My screen time stayed the same as usual."
Sentiment: Neutral

Comment: "I feel stressed when I see how much time I spend on social media."
Sentiment: Negative

Now classify the following comment:

Comment: "{user_comment}"
Sentiment:
"""

response = client.responses.create(
    model="gpt-4.1-mini",
    input=prompt,
    temperature=0.2,  # Low temperature for consistent classification
    max_output_tokens=16 # Changed from 10 to 16 to meet the minimum requirement
)

print("User Comment:", user_comment)
print("Predicted Sentiment:", response.output_text.strip())
```

```
User Comment: I feel stressed when I see how much time I spend on social media.
Predicted Sentiment: Negative
```